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# RV COLLEGE OF ENGINEERING®

**(An Autonomous Institution Affiliated to VTU)**

**V Semester B. E. Regular Examinations Feb/March -2025**

## Common to CSE/ISE

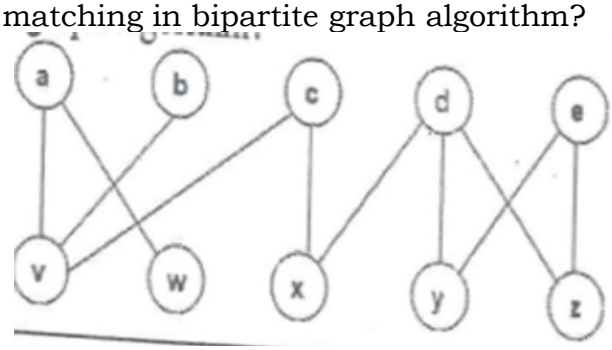
# ADVANCED ALGORITHMS

***Time: 03 Hours*** ***Maximum Marks: 100***

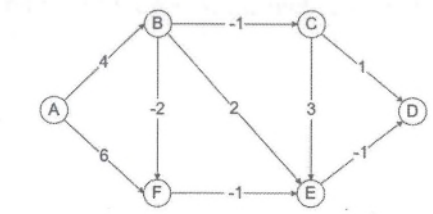
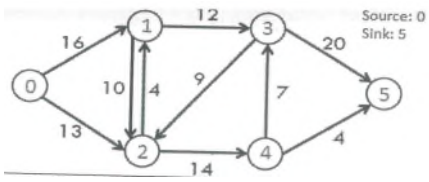
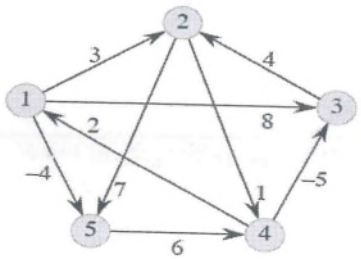
***Instructions to candidates:***

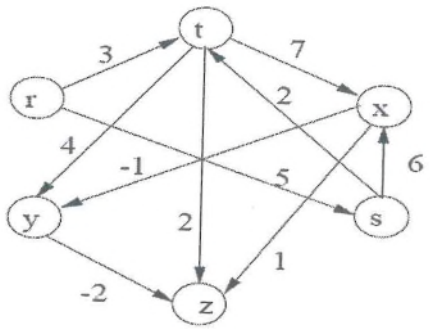
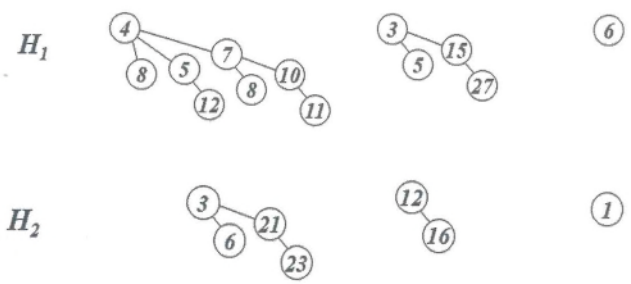
1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

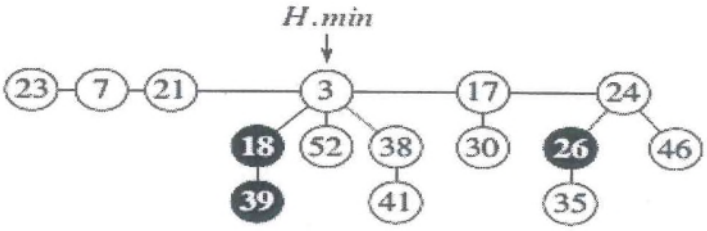
**PART-A**

| | | | M | BT | CO |
|---|---|---|---|---|---|
| 1 | 1.1 | What is recurrence? Write the solution for the recurrence. $T(n) = \begin{cases} 1 & if \quad n = 1, \\ 2T(n/2) + n & if \quad n \geq 1, \end{cases}$ | 02 | 2 | 1 |
| | 1.2 | Give a tight asymptotic upper bond on the solution for the following recurrence. $T(n) = 2T(n/8) + \sqrt[3]{n}$ | 01 | 2 | 1 |
| | 1.3 | The amortized cost of the $i^{th}$ operation with respect to potential function $\Theta$ is defined by _____. | 01 | 1 | 1 |
| | 1.4 | For the pattern $P = abaa$ in the text $T = abcabaabacabac$, find the valid shift $S$ and also the number of times the pattern occurs in the text. | 01 | 2 | 2 |
| | 1.5 | Consider a sequence $PUSH, POP$ and $MULTIPOP$ operations on an initially empty stack. Calculate the amortized cost of the above various operations using accounting method of analysis. | 02 | 3 | 2 |
| | 1.6 | _____ method follows a financial model to perform the amortized analysis. | 01 | 1 | 1 |
| | 1.7 | The residual capacity of an augmenting path $p$ is given by _____. | 01 | 1 | 1 |
| | 1.8 | From the given graph, how many vertices can be matched using maximum matching in bipartite graph algorithm?   | 01 | 1 | 1 |
| | 1.9 | Give the running time of all-pairs shortest path algorithm. | 01 | 1 | 1 |
| | 1.10 | _____ is a partition of the vertices of a graph in two disjoint subsets that are joined by atleast one edge. | 01 | 1 | 2 |
| | 1.11 | The composite numbers which fails Fermat's test is _____. | 01 | 2 | 2 |
| | 1.12 | What is the number of elements in $Z^*{}_{14}$ solve using Euler's phi function $\varphi(14)$. | 02 | 3 | 2 |
| | 1.13 | Compute the point-value representation for a given polynomial $8x^3 - 6x + 3$. | 02 | 3 | 3 |
| | 1.14 | The parallel $FFT$ circuit has depth of _____. | 01 | 2 | 2 |
| | 1.15 | Find the interpolating polynomial for the given is the data set $\{(0,1), (1,0), (2,5), (3,22)\}$. | 02 | 3 | 3 |

**PART-B**

| | | | | | |
|---|---|---|---|---|---|
| 2 | a | Solve the following recurrence using recursion tree method. Show the recursion tree and discuss how you obtained the solution from the tree. $T(n) = 4\,T\left(\frac{n}{2}\right) + n^2$ | 06 | 2 | 1 |
| | b | Solve the following recurrence using the Master Theorem, also state which case is applicable. $T(n) = 5T\left(\frac{n}{3}\right) + nlog\,(n)$ | 04 | 2 | 1 |
| | c | Suppose we perform a sequence of $n$ operations on a data structure in which $i^{th}$ operation costs $i$, if $i$ is an exact power of 2, and 1 otherwise. Use aggregate method of analysis to determine the amortized cost per operation. | 06 | 2 | 2 |
| | | | | | |
| 3 | a | Sort the given input $2, 5, 3, 0, 2, 3, 0, 3$ using Counting sort. Also show the stepwise calculations and the complexity of the algorithm at each level. | 08 | 3 | 2 |
| | b | With an algorithm, find the longest common subsequence of the given two sequences $X = \{\,A, B, A, A, B, A\,\}$ and $Y = \{B, A, B, B, A, B\}$. | 08 | 3 | 2 |
| | | **OR** | | | |
| 4 | a | Write an algorithm for sorting the array elements using Bucket Sort. Use the above algorithm to sort the given array $22, 45, 12, 8, 10, 6.72, 81, 33, 18, 50, 14$. Show the step wise sorting details. | 08 | 3 | 2 |
| | b | Write Dynamic programming-based algorithm to solve the matrix chain multiplication problem. Solve the given sequence $\{4, 10, 3, 12, 20$, and $7\}$. The matrices have size $4 \times 10, 10 \times 3$, $3 \times 12, 12 \times 20$, $20 \times 7$ | 08 | 3 | 2 |
| | | | | | |
| 5 | a | Demonstrate with an algorithm, the working procedure of bellman ford algorithm for solving the single source shortest path problem shown in Fig 5a.   Fig 5a   fig 5b | 08 | 3 | 4 |
| | b | Write and apply Ford Fulkerson algorithm to compute the maximum network flow for the given graph $G$ shown in fig 5b and $S$ as source and $T$ as sink nodes. | 08 | 3 | 3 |
| | | **OR** | | | |
| 6 | a | Use Johnson's algorithm to find the shortest paths between all pairs of vertices in the graph given below.   | 10 | 3 | 3 |

| | | | | | |
|---|---|---|---|---|---|
| | b | Run $DAG - SHORTEST-PATHS$ on the directed graph given below, using vertexs as the source.<br>  <br> | 06 | 3 | 3 |
| | | | | | |
| 7 | a | Solve the following simultaneous linear congruences using Chinese Remainder Theorem. $x = 2(mod5), x = 3(mod\ 7)$, $x = 10\ mod(11)$. | 05 | 3 | 2 |
| | b | Explain $RSA$ public-key cryptosystem. Consider an $RSA$ key set with $p = 11, q = 29$. What value of d should be used in the secret key? What is the encryption of the message $M = 100$? | 06 | 3 | 3 |
| | c | Write Knuth–Morris-Pratt algorithm to compute the prefix function and also compute the prefix function π for the pattern, $ababbabbabbababbabb$. | 05 | 3 | 4 |
| | | **OR** | | | |
| 8 | a | Find all solutions to the equations $x \equiv 2(mod3), x \equiv 3(mod\ 5)$, $x \equiv 2\ (mod7)$. | 05 | 3 | 3 |
| | b | Demonstrate with procedure, the extended form of Euclid's algorithm for the given equation $d = gcd(a, b) = ax + by$. Also illustrate how $EXTENDED - EUCLID$ computes $gcd(81,57)$. | 06 | 3 | 4 |
| | c | Write the algorithm to calculate the transition function and the finite automaton matcher for string matching with finite automata. Using the text string $T = abababacaba$, demonstrate the string-matching automaton's operations and build it for the pattern $p = ababaca$. | 05 | 3 | 3 |
| | | | | | |
| 9 | a | Draw the parallel $FFT$ circuit which computes the $FFT$ on $n$ inputs for $n = 8$. | 08 | 4 | 4 |
| | b | Consider two binomials queues, $H_1$ and $H_2$ given below.Explain with steps how to merge them.<br>  <br> | 08 | 3 | 3 |
| | | **OR** | | | |

| | | | | | |
|---|---|---|---|---|---|
| 10 | a | Define Fibonacci heap. Write an algorithm to extract minimum node from the Fibonacci heap. Using the above algorithm, delete the minimum key for the given graph and consolidate it.   | 08 | 4 | 4 |
| | b | Explain with examples the two ways of representing polynomials. | 08 | 3 | 2 |